MATH 238 # 25969 Finals Review

The Finals Exam will have problems on chapter sections **represented** below.
When solving ODEs **do not** use Laplace Transforms unless asked.

Test 1 Problems

Problem 1: Take the Initial Value Problem:

$$\frac{dy}{dx} + 3x^2y^2 = 0, \qquad y(1) = 4.$$

(a) The IVP has order (choose one) (i) 1 (ii) 2 (iii) 3
(b) The IVP is (choose one) (i) linear (ii) nonlinear.
(c) (Do not show working) The function $y(x) = \dfrac{4}{4x^2 - 3}$ is a solution (i) yes (ii) no.
(d) An appropriate method of getting the solution is _____.

(e) The differential equation in differential form is _____.

Problem 2: Take an ode $x'' - 2x' + 5x = 0$ with a solution $x(t) = 3e^t \sin(2t)$.

(a) The ode is (choose one) (i) linear homogeneous (ii) linear nonhomogeneous (iii) nonlinear.

(b) Verify the function $x(t)$ is indeed a solution to the ode.

(c) Give the initial condition at $t = 0$.

Problem 3: Consider the differential equation $y' = \sqrt{y^2 - 9}$. Let $f(x, y) = \sqrt{y^2 - 9}$.

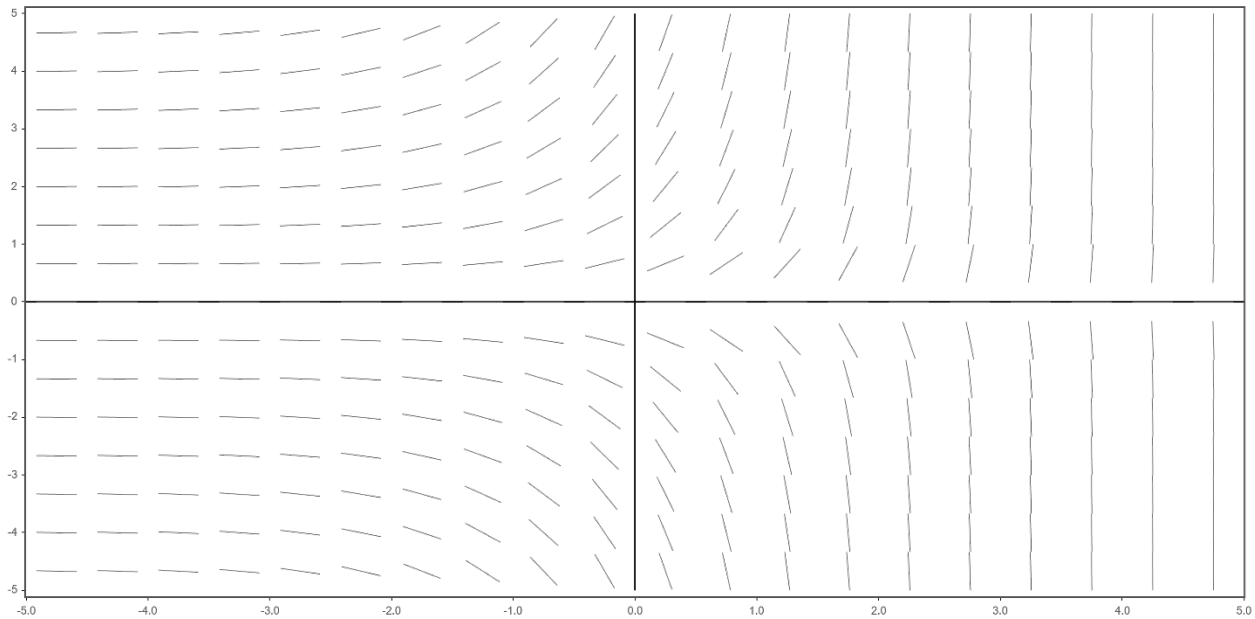
(a) Find the partial derivative $f_y(x, y)$.

(b) Give the two largest rectangles of xy plane in the form $(a,b) \times (c,d)$ for existence and uniqueness of the solution of the differential equation.

(c) Determine whether (yes/no) the existence/uniqueness Theorem guarantees that the differential equation possesses a unique solution through (4, 3). No working needed.

(d) Which rectangle contains the solution through the initial condition $y(0) = 4$?

Problem 4: The slope field below is for the ODE $\dfrac{dy}{dx} = ye^x$. Sketch two solutions with initial condition $y(0) = 1$ and $y(0) = -1$ respectively.

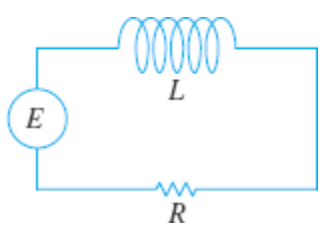


Problem 5: Consider the autonomous differential equation $dy/dx = y(y-1)^3$. Locate the critical points of the differential equation.

(a) Sketch a phase portrait of the differential equation.
(b) Sketch typical solution curves in the subregions in the xy-plane determined by the graphs of the equilibrium solutions.

Problem 6: Solve the IVP and give an explicit solution $y(x)$: $\dfrac{dy}{dx} = e^{5x+3y}$, $y(0) = 0$.

Problem 7: Determine the ODE in the current function $i(t)$ through the circuit.



Problem 8: Solve the ode: $\dfrac{dy}{dx} = \left(\dfrac{y+3}{x+5}\right)^2$.

Problem 9: Use the integrating factor method to solve the ode of a falling object with air resistance. $m\dfrac{dv}{dt} = mg - kv$.

Problem 10: Solve the IVP $2y' - y = 4\sin(3t)$, $y(0) = 3$. Use the integrating factor method.

Test 2 Problems

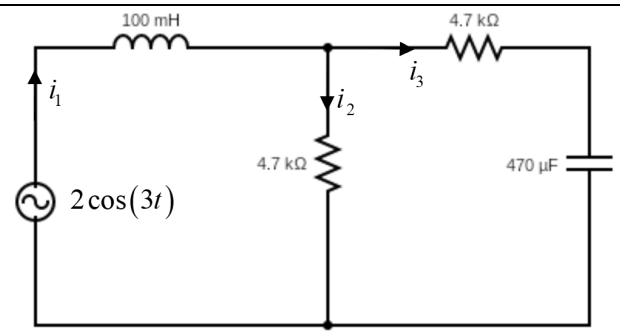
Problem 1: The following differential equation models population growth with harvesting.

$$\frac{dP}{dt} = kP - h, \quad P(0) = P_0$$

Solve the differential equation.

Problem 2: A breeder reactor converts relatively stable uranium-238 into the isotope plutonium-239. The isotope disintegrates. Suppose also c grams are added to the breeder every year. Set up an IVP for the amount at any time t, if an amount A_0 was observed **initially**.

Problem 3: Consider an electrical circuit labelled below with a variable voltage source with $V(t) = 2\cos(3t)$. Suppose one differential equation is given. Use Kirchoff's first and second equations to set up the second ordinary differential equation in $i_1(t)$ and $i_2(t)$.

	$L\frac{di_1}{dt} + R_1 i_2 = 2\cos(3t)$ where $L = 100\times10^{-3} H$ and $R_1 = 4.7\times10^3$ Ohms.

Problem 4: Solve the differential equation. $y''' - 2y'' + y' - 2y = 0$.

Problem 5: Use the undermined coefficient method to find the solution to $y'' - 6y' + 9y = 3xe^{3x}$.

Problem 6: Solve the differential equation. $x^2 y'' - 20y = 0$.

Problem 7: Solve the differential equation $y'' - 4y = x$ by variation of parameters.

Problem 8: Find the largest interval centered about $x = 0$ for which the given initial-value problem has a unique solution. (Enter your answer using interval notation.)

$$y'' + (\tan(x))y = e^x, \quad y(0) = 1, y'(0) = 0.$$

Problem 9: Use the Wronskian to test for linear independence in the first two sets. Use definition in the second two sets to test for linear independence.

$y_1 = e^{2x}, y_2 = xe^{2x}$	$y_1 = 2 + x, y_2 = x, y_3 = x^2$
$y_1 = 2, y_2 = 1 + x, y_3 = 3x$	$y_1 = 0, y_2 = x, y_3 = x^2$.

Problem 10: The function $y_1 = 1$ is a solution of the differential equation $\left(1-x^2\right)y''+2xy'=0$.

Use reduction of order or formula $y_2 = y_1\int \frac{e^{-\int Pdx}}{y_1^2}dx$ to find a second solution .

Other Problems [No solutions and answers provided]

Problem 1: A 500-liter tank initially contains 10 *g* of salt dissolved in 200 liters of water. Water that contains 1/4 g of salt per liter is poured into the tank at the rate of 4 liters/min and the mixture is drained from the tank at the rate of 2 liters/min. Find a differential equation for the quantity *Q(t)* of salt in the tank at time *t* prior to the time when the tank overflows. Solve the equation.

Problem 2: Find all the eigenvalues and eigenfunctions for the boundary value problem $y''+\lambda y=1, y'(0)=0, y'(2\pi)=0$ (consider just the case $\lambda < 0$).

Problem 3: Find the Laplace transforms by evaluating the integral $F(s)=\int_0^\infty e^{-st}f(t)\,dt$:

(a) t (b) te^{-t}	(c) $\sinh(bt)$ (d) $e^{2t}-3e^{t}$	(e) t^2 (f) $t\sin t$.

Problem 4: Use rules to write the Laplace transform of

(a) $\cosh t \sin t$ (b) $\sin^2 t+\cos^2 2t$	(c) $\sin(t+\pi/3)$ (d) $e^t \sin t$	(e) $t^3\cos t$ (f) t^3e^{2t}.

Problem 5: Find the Laplace transform of:

(a) $f(t)=\begin{cases} e^{2t}, & 0\le t<1 \\ e^{-2t}, & t\ge 1 \end{cases}$	(b) $f(t)=\begin{cases} 2, & 0\le t<4 \\ t, & t\ge 4 \end{cases}$.

Problem 6: Find the inverse Laplace transform:

(a) $\frac{3}{(s-7)^4}$ (b) $\frac{2s-3}{s^2-4s+13}$	(c) $\frac{s^2-1}{\left(s^2+1\right)^2}$ (d) $\frac{1}{s}-\frac{s}{s^2+1}$.

Problem 7: Solve the initial value problems using Laplace Transforms:

(a) $y''+3y'+2y=e^t, y(0)=1, y'(0)=-6$.

(b) $y''+y=\sin 2t, y(0)=0, y'(0)=1$.

(c) $2y''+2y'+y=2t, y(0)=1, y'(0)=-1$.

(d) $y''+y=f(t), y(0)=0, y'(0)=1$ where $f(t)=\begin{cases}0, & 0\le t<1\\ 2, & t\ge 1\end{cases}$.

Problem 8: Without solving the given differential equation, find the minimum radius of convergence of the power series about the ordinary point $x=0$. About the ordinary point $x=1$.

(a) $(x^2-25)y''+2xy'+y=0$.

(b) $(x^2-7x+10)y''+xy'-4y=0$.

(c) $(x^2-2x+10)y''+xy'-4y=0$.

Problem 9: Find two linearly power series solutions of the given differential equation about the ordinary point $x=0$.

(a) $y''+y=0$.

(b) $y''-y'=0$.

(c) $y''+xy=0$.

Problem 10: Solve Euler equations

(a) $x^2y''+xy'+4y=0$.

(b) $x^3y'''+xy'-y=0$.

Problem 11: Solve the Bernoulli equation

(a) $x\dfrac{dy}{dx}+y=x^2y^2$.

(b) $x\dfrac{dy}{dx}+y=x^2y^3$.